A Systematic Literature Review of LLM-Based Linguistic Steganography: Contextual Compatibility, Evaluation challenges, and Trade-offs

NASOUH ALOLABI AND RIAD SONBOL, Higher Institute for Applied Sciences and Technology, Syria

Steganography is the practice of concealing information within seemingly ordinary text. The rapid growth of Large Language Models (LLMs) has led to increasing research attention on their potential for linguistic steganography. This systematic literature review examines how LLMs are reshaping this field through 34 accepted papers published between 2018 and 2025. Through analysis of primary studies, the review synthesizes recent work on LLM-based steganography, with particular attention to application domains, evaluation practices, external knowledge integration, and recurring technical trade-offs. The surveyed literature suggests that LLM-based methods can improve text fluency and contextual plausibility relative to earlier linguistic steganography techniques, but it also shows that gains in payload, naturalness, and security remain difficult to balance in practice. A recurring theme across the reviewed studies is the growing importance of contextual compatibility, meaning the extent to which generated cover text fits the surrounding discourse, task, and conversational setting rather than merely sounding fluent. This includes domain adaptation, retrieval, and semantic guidance as mechanisms for improving both usability and concealment. The review concludes by identifying ongoing challenges.

Additional Key Words and Phrases: Systematic Literature Review, Linguistic Steganography, Large Language Models, LLMs, Natural Language Processing, NLP, Black-box Steganography, Context Retrieval, Generative Text Steganography, Imperceptibility

Preprint Notice: This is a preprint version of our systematic literature review, last updated on May 23, 2026. The manuscript is currently under review.

1 INTRODUCTION

Steganography hides a secret message inside ordinary-looking content, such as an image, an audio clip, or text, so that the communication is not readily apparent [12, 38]. When the cover is text, the broader field is *text steganography*. It includes format-based techniques [12]. When the method relies on natural-language content rather than formatting alone, it is called *linguistic* steganography [12, 56]. Large language models (LLMs) have become an important component in many such methods. This motivates the synthesis in this paper [2, 23, 39, 47, 48].

In linguistic steganography, redundancy is limited [16, 53]. There are few harmless ways to change wording without changing meaning or sounding unnatural. This makes concealment fragile. A single awkward synonym, unusual phrase, or implausible sentence can expose hidden communication. Classical lexical and syntactic methods therefore tend to provide low payloads. They also leave artifacts that modern steganalysis can exploit [20, 45, 54].

These models have changed this landscape by making high-quality text generation widely accessible [2, 3, 48]. Their fluency, contextual sensitivity, and broad domain coverage have enabled a new generation of steganographic methods. These methods rely on text generation, paraphrasing, retrieval, and prompting, rather than only local word substitution [20, 23, 39, 41, 47].

These advances do not remove the core trade-offs of the field. Increasing embedding capacity can still degrade fluency, coherence, or statistical naturalness. Optimizing for stealth may also reduce throughput or robustness [15, 16, 27, 53]. This tension is often described as the Perceptual-Statistical Imperceptibility Conflict (PSIC) [53]. A systematic synthesis is needed to clarify how recent work defines the problem, evaluates success, and balances practical constraints. Several

Author's address: Nasouh AlOlabi and Riad Sonbol , Higher Institute for Applied Sciences and Technology, Damascus, Syria.

surveys and related reviews already discuss text or linguistic steganography [25, 36, 45]. However, they do not foreground *contextual compatibility* as a central criterion for assessing LLM-based methods.

This review addresses that gap by synthesizing recent primary studies on LLM-based linguistic steganography through the lens of contextual compatibility. We do not treat generated text as acceptable just because it is fluent or statistically plausible. Instead, we examine how closely it fits the surrounding discourse, task, and conversational setting. This paper maps the emerging design space. It also highlights recurring evaluation practices and the main unresolved limitations.

The rest of the paper is structured as follows. Section 2 introduces the background on steganography, its evaluation criteria, and the capabilities of LLMs. It also covers their role in generative linguistic steganography. Section 3 reviews related surveys to position the present study. Section 4 outlines the review methodology, including the research questions, search strategy, and selection criteria. Section 5 presents the findings across five research questions (RQ1–RQ5). Section 6 discusses the implications and limitations of the review. Section 7 concludes the paper and outlines future research directions.

2 BACKGROUND

Information security systems broadly encompass **encryption**, **privacy**, and **concealment**, the last of which -known as **steganography**- is the focus of this review. While encryption and privacy protect message content, they do not conceal the existence of communication, which may itself arouse suspicion. Steganography instead prioritizes **imperceptibility**: embedding information into ordinary carriers (e.g., images or text) so that hidden messages remain unnoticed.

Text is a particularly challenging carrier due to its low redundancy and strict semantic constraints. The classical "Prisoners' Problem" [38] illustrates the goal: two parties, Alice and Bob, must exchange hidden information without alerting a watchful adversary.

Textual steganography methods are typically divided into **format-based** approaches, which exploit layout or structural features, and **linguistic-based** approaches, which modify linguistic form. Within the latter, early techniques such as **synonym substitution** embed bits by altering lexical choices, but suffer from low capacity and high detectability.

More formally, generative linguistic steganography operates through an **embedding function** f_{emb} that establishes a cryptographic and linguistic mapping to hide secret messages within generated text. The algorithm takes as inputs a language model $\mathcal{L}$ and a secret message m (represented as a uniformly distributed bitstream). At each generation step, rather than sampling freely from the language model's conditional distribution p_{LM}, the embedding function f_{emb} **restricts or modifies** the token selection process according to the message bits being embedded. This specialized process is known as **steganographic sampling**, producing a steganographic text (stegotext) y. The security of the system depends on minimizing the divergence between the original language model distribution p_{LM} and the modified distribution q induced by f_{emb} [57].

Traditional linguistic approaches offer limited embedding capacity and often leave statistical artifacts. Advances in deep learning and **Large Language Models (LLMs)** now enable generative methods that achieve higher text quality and more secure embedding. Evaluating such systems requires several dimensions of imperceptibility: **perceptual** (human naturalness), **statistical** (distributional similarity to natural text), and **cognitive** (semantic and contextual fidelity) [8].

However, the use of LLMs also introduces new sources of detectability. One important issue is **hallucination**, where a model generates content that is factually unsupported, semantically inconsistent, or weakly grounded in the preceding context. In ordinary text generation, hallucinations are often discussed as errors of reliability. In steganographic generation, however, they also become a security concern: even if a sentence appears fluent at the surface level, an unsupported claim, abrupt topic shift, or contradiction with the surrounding context may make the stegotext suspicious. Hallucination therefore primarily threatens **cognitive imperceptibility**, because the generated text must not only sound natural, but also remain contextually plausible and communicatively coherent.

A second challenge is the **PSIC effect** (Perceptual-Statistical Imperceptibility Conflict) [53], which captures a central tension in generative linguistic steganography. Text that is optimized for human fluency is not necessarily statistically indistinguishable from natural human text. For example, selecting only high-probability tokens can produce sentences with low perplexity and strong local readability, but it may also make the generated distribution overly smooth, regular, or concentrated compared with real human writing. Such text can appear natural to readers while remaining detectable to statistical steganalyzers. [53]

The PSIC effect therefore shows that **perceptual imperceptibility** and **statistical imperceptibility** are related but not identical goals. Human-written text often contains variation, irregularity, and locally unexpected word choices. By contrast, overly fluent model-generated text may lack this natural variance. Increasing the embedding rate can sometimes force the sampler to select from a broader range of tokens, which may better approximate the diversity of human text; however, this same process can also reduce readability or semantic coherence if the token choices become poorly aligned with the context. Thus, higher capacity does not automatically imply better security. Secure generative steganography requires balancing embedding rate, fluency, distributional similarity, and contextual consistency.

2.1 Capabilities and Approximating Natural Communication

Large Language Models (LLMs) are autoregressive, generative systems based on the Transformer architecture [40] that approximate high-dimensional distributions over natural-language sequences [16][33]. Given a prefix, an LLM emits a probability vector over the vocabulary; the next token is sampled from this vector and appended to the prefix, and the process repeats until a stopping criterion is met. During pre-training, billions of parameters are tuned on large text corpora so that the model's predictive distribution converges to the empirical distribution of the data [2]. As a consequence, modern LLMs routinely produce text whose fluency, coherence and style are indistinguishable from human writing [3]. The learned latent representations capture stylistic and semantic regularities that generalize across domains, enabling applications requiring nuanced linguistic mimicry [35].

2.2 Role in Generative Linguistic Steganography

LLMs are considered **favorable for generative text steganography** due to their ability to generate high-quality text. Researchers propose using generative models as steganographic samplers to embed messages into realistic communication distributions, such as text. This approach marks a departure from prior steganographic work, motivated by the public availability of high-quality models and significant efficiency gains.

LLMs like **GPT-2** [33], **LLaMA** [48], and **Baichuan2** [52] are commonly used as basic generative models for steganography. Existing methods often utilize a language model and steganographic mapping, where secret messages are embedded by establishing a mapping between binary bits and the sampling probability of words within the training vocabulary. However, traditional "white-box" methods necessitate sharing the exact language model and training vocabulary, which limits fluency, logic, and diversity compared to texts generated by modern black-box LLM APIs. These methods inevitably alter sampling probability distributions when embedding messages, posing security risks [47]. **Black-box approaches** avoid these limitations by leveraging state-of-the-art hosted models to deliver better text quality, faster generation speeds, and minimal local resource requirements. However, this paradigm trades fine-grained control over internal sampling for these practical advantages. Conversely, **white-box access** enables precise parameter control and stronger security guarantees, but demands substantial computational resources, engineering effort, and higher latency—raising deployment barriers. This trade-off between access paradigms is central to evaluating the robustness and applicability of modern linguistic steganography.

New approaches, such as **LLM-Stega** [47], explore **black-box generative text steganography using the user interfaces (UIs) of LLMs**. This circumvents the requirement to access internal sampling distributions. The method constructs a keyword set and employs an encrypted steganographic mapping for embedding. It proposes an optimization mechanism based on reject sampling for accurate extraction and rich semantics [47].

Another framework, **Co-Stega**, leverages LLMs to address the challenge of low capacity in social media. It expands the text space for hiding messages through context retrieval and **increases the generated text's entropy via specific prompts** to enhance embedding capacity. This approach also aims to maintain text quality, fluency, and relevance [22].

The concept of **zero-shot linguistic steganography** with LLMs utilizes in-context learning, where samples of covertext are used as context to generate more intelligible stegotext using a question-answer (QA) paradigm [23]. LLMs are also employed in approaches like **ALiSa**, which directly conceals token-level secret messages in seemingly natural steganographic text generated by off-the-shelf BERT [7] models equipped with Gibbs sampling [54].

The increasing popularity of deep generative models has made it feasible for provably secure steganography to be applied in real-world scenarios, as they fulfill requirements for perfect samplers and explicit data distributions (see Section 2) [10, 16, 31].

LLM-based steganographic methods are typically evaluated on two primary axes: imperceptibility (perceptual, statistical, and cognitive measures) and embedding capacity (e.g., bits per token or bits per word). Imperceptibility evaluations may include automatic metrics (PPL, Distinct-n, MAUVE, KL/JSD) as well as human judgements; embedding capacity is usually reported as bits/token or overall embedding rate.

3 RELATED REVIEWS

Majeed et al. (2021) [25] surveyed pre-LLM text steganography techniques, predating the current transformer era. Setiadi et al. (2025) [36] recognizes that LLMs have "revitalized" linguistic steganography, examining recent methods (2021-2025) using GPT-2 [34], GPT-3 [2], LLaMA2 [**?**], and Baichuan2 [50]. However, their papers are best described as surveys or reviews, but they do not foreground contextual compatibility or LLM-focused approaches as central criteria, and they are not exhaustive systematic literature reviews. More importantly, neither review foregrounds *contextual compatibility* as a core criterion for judging whether generated cover text truly fits its surrounding discourse, task, and conversational setting. Crucially, the field has evolved from "statistical vector embedding" (Word2Vec, GloVe) to "language-model vector embedding" that exploits BERT-scale transformers and higher-dimensional semantic spaces.

This creates a methodological gap: no systematic review comprehensively maps how large-scale transformers have redefined text steganography. Modern advances extend beyond naive generation to sophisticated Controllable Text Generation (CTG) frameworks [56]. These employ Variational Autoencoders (VAEs) to model latent features and Diffusion Models to inject randomness, mitigating spurious associations between secrets and control conditions. Classical surveys emphasized synonym replacement, spacing manipulation, and Huffman coding [25]-techniques that predated LLMs. Earlier methods relied on context-free grammars (CFGs) or Markov chains, often producing syntactically correct but semantically incoherent cover texts. Contemporary approaches leverage prompt learning and prefix tuning, enabling efficient model customization without costly full fine-tuning.

Defensive strategies must evolve accordingly. Traditional steganalysis, premised on hand-crafted statistical features, falters against generative steganography's high statistical concealment. Current research must confront "stegomalware"-attacks that conceal command-and-control communications within innocuous digital media.

4 RESEARCH METHOD

This study was undertaken as a systematic literature review using the guidelines presented in Petersen et al. [30]. The goal of this review is to identify, categorize, and analyze existing literature published between 2018 and 2025 on LLM-based linguistic steganography, with particular attention to context handling, evaluation practice, and unresolved technical limitations.

4.1 Planning

In this section, we define the research questions, the search strategy, and the inclusion and exclusion criteria used to filter the retrieved studies.

4.1.1 Research Questions. This systematic literature review is guided by the following five research questions:

- **RQ1:** What is the current state of published literature on LLM-based steganography?
- **RQ2:** What applications are being explored for LLM-based steganographic techniques?
- **RQ3:** What evaluation metrics and methods are used to assess LLM-based steganography?
- **RQ4:** How are external knowledge sources integrated into LLM-based steganographic systems?
- **RQ5:** What limitations and trade-offs characterize current LLM-based steganographic techniques?

RQ1 provides a general overview of the published literature. RQ2 identifies the main application areas addressed by the selected studies. RQ3 examines how LLM-based steganographic methods are evaluated. RQ4 focuses on the role of external knowledge and contextual information. RQ5 summarizes the limitations, trade-offs, and open challenges reported across the literature.

4.1.2 Search Strategies. The initial literature search used the following query string:

> (steganography OR watermark OR "Information Hiding") AND ("Large Language Model" OR LLM OR BERT OR LAMA OR GPT)

The query was intentionally broad to maximize recall during the identification stage, even though the final synthesis focuses on LLM-based linguistic steganography. It was executed across ACM Digital Library, IEEE Digital Library, Science@Direct, Scopus, and Springer Link. To complement the automated search, backward snowballing was applied by examining the reference lists of included studies. Snowballing mainly retrieved older linguistic steganography studies that did not explicitly mention LLMs but provided useful methodological context for interpreting recent work.

4.1.3 Inclusion and Exclusion Criteria. To ensure the selection of high-quality and relevant studies, the following criteria were applied.

Inclusion Criteria Studies were included if they:

IC1: Provided full-text access.
IC2: Were published in English from 2018 onwards.
IC3: Appeared in peer-reviewed journals, conferences, or workshops.

IC4: Directly addressed linguistic steganography or closely related information-hiding techniques involving or significantly impacted by LLMs, BERT, LLaMA, or GPT architectures.
IC5: Represented empirical studies, surveys, reviews, or theoretical contributions.

Exclusion Criteria Studies were excluded if they:

EC1: Were duplicates, retaining the most complete or recent version.
EC2: Were incomplete, abstract-only, or irrelevant to steganography with LLMs.
EC3: Were non-English publications.
EC4: Came from non-peer-reviewed sources such as preprints, dissertations, theses, books, or book chapters, unless extended from peer-reviewed conference papers.

4.2 Conducting the Search

The initial automated search across the selected digital libraries yielded a total of 1043 candidate papers. The distribution by source was ACM Digital Library (346), IEEE Digital Library (61), Science@Direct (209), Scopus (151), and Springer Link (276). Consistent with the scope of this review, only studies published between 2018 and 2025 were considered during screening and selection.

Duplicate papers were eliminated using the Parsifal tool[1] before the remaining papers underwent a multi-stage filtering process based on title, abstract, and full-text review. After title and abstract filtering, 58 papers remained. Of these, 34 were accepted for analysis with usable bibliographic and extraction data, while 6 additional papers were identified as potentially relevant but remained pending full extraction at the time of analysis.

4.3 Data Extraction and Classification

A Data Extraction Form (DEF) was developed to systematically collect data from each primary study and answer the research questions. The form was organized as a table with the following types of information:

- Bibliometric information: paper title, type (steganography or watermarking), author(s), publication year, and publication venue.
- Model details: input and output formats, key characteristics, approach classification, specific LLM used, embedding process description, and code availability.
- Datasets: all datasets employed, including their sizes when reported.
- Context awareness: whether the method is explicit, implicit, or absent; the context keyword; how context is represented; and how it is used in the method.
- Evaluation details: evaluation metrics, steganalysis models used, and the best numerical results for each reported metric.
- Strengths and limitations: the main strengths and weaknesses of the approach or model.

Following data extraction, studies were classified using predefined categories derived from the research questions. The results are summarized using tables and descriptive statistics, and each research question is addressed individually with interpretation of findings and identification of future research directions.

[1] https://parsif.al

5 RESULTS

This section presents and discusses the results obtained from the analysis of 34 primary studies on LLM-based steganography. The findings are organized according to the five research questions defined in Section 4. For each question, we summarize the observed trends and interpret what they indicate about the current state of the field.

5.1 State of Published Literature on LLM-based Steganography (RQ1)

5.1.1 Publication Trends and Distribution. Our analysis of 32 studies indicates a clear rise in LLM-based steganography research after 2023, with most of the reviewed studies appearing in 2024–2025. This increase coincides with the rapid diffusion of large-scale language models in both proprietary and open-weight forms, including widely used families such as GPT, GPT-2, and LLaMA [2, 34**?**]. Across the reviewed corpus, open-weight and research-accessible models appear more frequently than closed commercial systems, reflecting the practical importance of reproducibility and controllable experimentation.

The field has also evolved from earlier white-box token-sampling schemes toward a broader design space that includes hybrid and black-box methods. The reviewed studies now span generative schemes that write stego text from scratch [10, 16, 53, 57], rewriting systems that paraphrase existing covers [20, 28], black-box pipelines that treat the model as an opaque API [18, 39, 46, 47], zero-shot protocols driven primarily by prompting [23], context-aware frameworks that retrieve or constrain external information to increase semantic plausibility or payload [22, 37, 41], and methods that emphasize stronger security, robustness, or token-level quality control [6, 10, 16, 24, 43].

Table 1 summarizes the year-wise publication growth, while Table 2 shows how the reviewed studies distribute across model families.

Table 2 gives the model-family breakdown used in the synthesis.

Table 1. Publication trends by year

Year	2020	2021	2022	2023	2024	2025	Total
Publications	2	1	2	5	14	8	32

Table 2. Representative model families across the 34 studies explicitly enumerated in the current synthesis tables. Categories are non-exclusive, so totals exceed 100%.

Model Type (Studies)	Models and Representative Works
Open-weight Models (26/34, 76.5%)	GPT-2 [10, 16, 29, 41], LLaMA/LLaMA2 [21–23, 31], BERT [8, 9, 15, 32, 49, 54–56, 58], OPT [27], BART [20, 32], Qwen [19, 37], XLNet [5]
Proprietary Models (7/34, 20.6%)	GPT-3.5/4, ChatGPT [13, 18, 39, 46, 47, 51]
Custom or Task-Specific Architectures (6/34, 17.6%)	From-scratch or task-specific models [6, 9, 15, 27, 43, 53]

5.2 Applications of LLM-based Steganographic Techniques (RQ2)

The reviewed studies show that **covert communication** remains the dominant application of LLM-based linguistic steganography. In this setting, secret information is embedded into seemingly benign text so that the message can pass through ordinary communication channels with minimal suspicion. LLM-based methods improve this use case by producing more fluent and contextually plausible outputs than earlier rule-based or local-substitution systems [16, 47, 53].

Within this application space, the reviewed papers explore several operational variants. Some methods generate stego text from scratch using a shared model or sampling strategy [5, 10, 16, 43]. Others rely on paraphrasing or rewriting so that a pre-existing cover text can be transformed while preserving its apparent meaning [20, 28]. Recent black-box and prompt-based approaches such as **LLM-Stega**, **DeepStego**, and multi-criteria linguistic optimization show that covert communication can also be implemented through commercial or hosted interfaces without direct access to token probabilities, using prompt design, keyword sets, stylistic features, and reject-sampling style verification instead [18, 46, 47]. Context-aware systems such as **Co-Stega**, **Hi-Stega**, and emotionally controllable steganography further adapt the generated text to specific domains, topics, entities, emotional tones, or social settings in order to improve plausibility and increase effective payload [22, 37, 41].

Although adjacent topics such as watermarking, authorship attribution, or prompt-hiding attacks are relevant to the broader landscape of information hiding, they remain analytically distinct from covert payload transmission. The new 2025 watermarking studies are therefore treated as boundary cases: useful for understanding robustness, semantic preservation, and detection methods, but not interchangeable with steganographic communication systems [42].

A comprehensive breakdown of how these reviewed studies are distributed across these primary application areas and variants is summarized in Table 3.

Table 3. Distribution of Applications for LLM-Based Steganography

Primary Application	Variant / Category	Representative Studies	Total Count
Covert Communication	Generation-based (from scratch)	[8, 10, 16, 18, 20, 23, 37, 42, 47, 53]	10
	Paraphrasing or Rewriting	[5, 6, 24, 28, 46]	5
	Black-box and Prompt-based	[15, 21, 49, 51, 54]	5
Context-Aware Systems	Domain, Emotion, or Social Adaptation	[19, 22, 27, 29, 41, 55, 58]	7
Watermarking	Content Attribution (Boundary Cases)	[9, 25, 31, 32, 39, 56]	6
		Total	**33**

5.3 Evaluation Metrics and Methods (RQ3)

Performance evaluation for LLM-based steganography relies on three key categories of metrics, with significant variation in reporting standards across studies. The analysis reveals both the diversity of evaluation approaches and the need for standardization.

5.3.1 Perplexity (PPL). An imperceptibility metric [14] that measures fluency, with lower values indicating better naturalness. It is recognized as a sensitive and unreliable metric for language model evaluation due to several intrinsic limitations. First, it suffers from a "confidently wrong" problem: as Baeldung, et al. [44] notes, perplexity measures only internal consistency, allowing models to assign low perplexity to grammatically perfect but factually absurd statements like "The cat is on the ceiling," since it cannot assess truth or logic. Second, it exhibits a short-text bias as Fang, et al. [11] demonstrated that perplexity scores are artificially inflated for short sequences despite potentially higher fluency, making it an "unqualified referee" for fair evaluation. Third, comparability across models is impossible without identical tokenization, vocabulary size directly scales perplexity - a model with fewer tokens appears deceptively better [26]. Fourth, perplexity fails to capture long-range dependencies in modern LLMs; Fang, et al. [11] argue that averaging log-likelihood across all tokens obscures performance on crucial "key tokens" by favoring predictable filler words. Finally, the metric is easily gamed through repetition, Wang, et al. [44] finds that "perplexity cannot distinguish between right emphasis and abnormal repetition," rewarding redundant text with artificially low scores. These flaws-sensitivity to length, architectural incompatibility, semantic blindness, and exploitability-collectively render perplexity an inadequate benchmark for steganographic text quality assessment.

5.3.2 *MAUVE.*

Another imperceptibility metric that evaluates distributional similarity between generated and reference text by quantifying the gap between neural and human-authored text using divergence frontiers. In this review, MAUVE appears only in a small minority of studies, and its interpretation is most natural when the authentic reference text is not itself produced by a large pretrained generator.

That makes MAUVE useful mainly as an *internal benchmark* for comparing variants within a single study. Reported scores are also *not directly comparable* across papers because scaling conventions, model architectures, and datasets differ substantially.

5.3.3 *Statistical Metrics.*

Kullback-Leibler Divergence (KLD) [17] and Jensen-Shannon Divergence (JSD) are information-theoretic metrics used to evaluate steganographic security. KLD quantifies information loss by measuring the relative entropy between cover and stego distributions, serving as the theoretical standard for security modeling despite being asymmetric and failing as a strict distance measure. JSD improves upon this as a symmetric, bounded variant that measures how far each distribution lies from their average, providing a more stable basis for formulating statistical imperceptibility bounds-particularly when language models approximate human text distributions. Together, these two attempt to capture how closely steganographic outputs mimic legitimate communication channels.

However, real-world application reveals critical reliability failures, most notably the Perceptual-Statistical Imperceptibility Conflict (PSIC effect). KLD and JSD scores increasingly diverge from human judgment as statistical optimization progresses: methods achieving superior divergence metrics often produce chaotic, low-quality text easily detected by human observers. This discrepancy manifests acutely in dataset dependency-identical methods yield KLDs of 19.507 on IMDB versus 8.295 on Twitter at equivalent embedding rates, rendering cross-paper comparisons meaningless. Further compounding this, researchers employ incompatible formulas (some using latent BERT features versus direct word distributions), feature spaces, and measurement scales, evidenced by Meteor's KLD ranging from 0.045 in one study to 7.491-11.845 in others. Consequently, these metrics function like rulers measuring paintings: they confirm technical dimensional accuracy while completely missing perceptual naturalness, necessitating parallel evaluation with human-centric measures to achieve genuine security.

5.3.4 *Capacity Metrics.*

Capacity is judged by four metrics:

- **Bits per Token (BPT):**
$$\text{BPT} = \frac{\text{Total Secret Bits}}{\text{Total Tokens}} \tag{1}$$
- **Bits per Word (BPW):**
$$\text{BPW} = \frac{\text{Total Secret Bits}}{\text{Total Words}} \tag{2}$$
- **Embedding Rate (ER) [4]:** Average density of hidden information per textual unit
$$\text{ER} = \frac{1}{N}\sum_{i=1}^{N} \text{bits}_i \tag{3}$$

where N is the number of textual units (words, tokens, or sentences) and bits_i is the number of bits embedded in the i-th unit. In some fixed-payload schemes, the total number of embedded bits is predetermined for each stego text and does not depend on its length. For example, LLM-Stega can assign a constant payload to every generated stego sentence, so the embedding rate can also be expressed as

$$\text{ER}_{\text{fixed}} = B_0 \tag{4}$$

where B_0 is the fixed number of bits embedded in each stego text.

- **Utilization Rate:**

$$H = -\sum_{x \in X} P(x) \log_2 P(x) \tag{5}$$

$$\text{UR} = \left(\frac{\text{Actual Bits Embedded}}{H} \right) \times 100\% \tag{6}$$

These quantities quantify how densely a secret is packed, yet they are riddled with systematic biases that invalidate cross-system comparison.

Table 4 summarizes the main bias sources that affect capacity reporting.

Table 4. Five primary bias categories affecting capacity metrics in steganographic evaluation

Bias Category	Core Problem	Critical Implication
Tokenization Inconsistencies	Metrics depend entirely on specific tokenizers (e.g., GPT-2 BPE vs. word-level)	Direct comparisons across papers become meaningless when tokenization strategies differ
The "PSIC Effect"	Conflicts between imperceptibility and statistical security are ignored	High capacity may degrade human fluency while paradoxically improving detection resistance
Model Training Bias	Utilization Rate calculations assume uniform token availability	Actual hiding space is smaller than theoretical entropy due to model frequency preferences
Reporting Ambiguities	No standard definition of "capacity" across systems	Practice payload vs. effective payload distinctions create misleading efficiency claims
Context Blindness	Density metrics treat text as neutral bit containers	Semantic incoherence constitutes a security failure that BPW/BPT fails to penalize

Tokenization differences make "1 BPT" from one paper incomparable to "1 BPT" from another due to the use of different tokenizers. The PSIC effect shows that higher density can hurt human fluency yet help statistical evasion. Model frequency preferences shrink the real alphabet to high-probability tokens, so naive entropy limits overstate usable space. Ambiguous reporting-practice vs. effective payload, ER1 vs. ER2, Bit Length vs. Stego Length-lets authors cherry-pick flattering numbers. Finally, BPW/BPT ignore semantics, rewarding gibberish that is obviously steganographic.

Recent works reveal additional distortions: loop-count overhead, dictionary-size caps, baseline-dependent "improvements," watermarking goals that invert the desired signal, conversational filler that dilutes BPW, and nonlinear ER curves that make any single threshold misleading.

The added 2025 studies reinforce this point. DeepStego emphasizes high capacity and recovery accuracy, multi-criteria optimization reports reliable stylistic decoding, and RS-Stega reports simultaneous PPL reduction and BPW improvement through token-level quality control [18, 43, 46]. These results are promising, but they also make cross-paper comparison harder because the reported gains depend on different baselines, prompts, datasets, and decoding assumptions.

5.3.5 Practical Capacity Analysis. To contextualize theoretical capacity metrics, Table 6 summarizes the embedding capacity across the reviewed methods using specific generated samples. Capacity calculations vary significantly based on the encoding mechanism. **Fixed-length mapping** techniques, such as **LLM-Stega** and **GCStego**, derive capacity from rigid keyword or token mappings—64 bits per sentence and 16 bits per word, respectively. **ParaLS** operates similarly but on a smaller scale, encoding a simple 3-bit binary watermark (“110”).

In contrast, **payload-dependent** methods scale with the cover text or secret message length. **Meteor’s** capacity of 1280 bits corresponds to an embedded payload of 160 bytes (e.g., Lorem Ipsum text). **Zero-shot** capacity is directly tied to the Base64 representation of the secret bitstream (encoding 24 bits per 4 characters). Finally, **Discop** utilizes a statistical approach; with a truncation parameter of $p = 0.95$, it achieves an entropy rate of 4.84 bits per token, yielding 968 bits over a 200-token sequence.

Table 5. Examples of stego texts. Limited by the space, we omitted the padding part.

24-bit Secret Message : (1,0,0,0,1,0,0,1,0,1,1,0,1,0,1,0,1,1,1,1,1,0,1,1)		
Cover Text	**Method**	**Stego Text**
sometimes we bring the story to you, sometimes you have to go to the story.	Rewriting-Stego $\text{BPT}_S = 1$	sometimes we bring it to your mind but sometimes you have to go back to it to find the story you want to tell...
	Rewriting-Stego $\text{BPT}_S = 2$	but we have a different way of thinking about what the story...
	Rewriting-Stego $\text{BPT}_S = 4$	some of all the story lines...
	Bins $\text{BPT}_S = 4$	somewhere and we have lots different...
	Masked-Stega ($\text{BPT}_S = 0.16$) (Encrypted message :10001)	sometimes we read the story to you , sometimes you have to stick to the story...

Table 6 provides a detailed catalog of stego text samples generated by the reviewed frameworks. It serves as a practical reference for the diverse capacities and calculation methods discussed previously, offering a direct comparison between the textual output and the numerical efficiency claimed by each study. This breakdown highlights the heterogeneity in how different researchers quantify the embedding density and its relation to the length of the generated cover.

Table 5 gives representative stego-text samples.

Table 6 lists the corresponding capacity calculations and source methods.

Table 6. Practical embedding capacity and calculation basis for selected stego text samples.

Stego Text	**Cap**	**Source**	**Calculation Basis**
"The picture in The Pale I HCR scientists' discussion now spans three dimensions. The first importance of the Yucatan Peninsula is demonstrated with the following conclusion: the Pliocene Earth has lost about seven times as much vegetation as the Jurassic in regular parts of the globe, from northern India to Siberia. Even since 1976, the continent has received or is already experiencing the worst mass extinction in recorded history since there were a million years last left from the extinction of Antarctic ice and the rapid spread of hydrocarbon-rich water as the Earth entered the Cretaceous Period. In turn, the world's oceans have been breathtakingly transformed and as a result the surrounding environments are especially vulnerable. The reason is obvious: there was no geological record of the presence of the Yucatan Peninsula in the Late Miocene epoch, which is the line that runs from the southern edge of South America to Siberia. The commander-in-chief of the Yucatan Peninsula, Frederick Nystrom, has determined that the area is to stay as barren as possible. As a result, individuals living on lake beds and on coastal areas have witnessed the loss of about 90 percent of their habitat. The Yucatan Peninsula consists of four zones, with two different habitats separated, each of which has experienced inseam damage. In one zone, along the northern shore of Lake Shemal, the retreating Tarahumara Ocean has been melted into a deep, seafloor called Nova Ravine, which south-east of the Yucatan Peninsula flows into Lake Isthmus, where there is an abundance of turtle life. A second beach, which lies at the far end of the peninsula, has been spewed down by a sea wall supporting Madagascar's Great Ocean Earthquake, 9,000 feet in magnitude and caused large numbers of deaths. The third zone, along the coast of Cancun and in Asuncion, is less severely affected. An estimated 16 percent of the continent is protected from the destruction of oceanic winds and floods. All three zones are in a state of catastrophic destruction. According to the definition provided by the National Commission on the extinction of the dinosaurs (Infection and Immunization in the Ind"	1280 bits	Meteor	160-byte payload × 8 bits
"Washington was martyred in the battle of Cullman in 1788. Although down 19 men, yet his mission accomplished. He ended his life in 1896. When he died of disease, three of the men responsible said that Washington had been moved and was seen at the funeral placing him in his father's arms. Another said that Washington treated his wounds with "a study of a personal sense of injustice." (A 16th century German historian calls Washington a liar because of this website.) He was also a friend of Dr. Henry H. Jackson, who was notable as George Washington's physician and a collaborator, and recognizes him by name in the any books about him. The last surviving manuscript is from 1888 and contains a frank and truthful account of the Quakers' plight. One story states that while fighting in Whitesburg, Washington succumbed to pneumonia. He was 38 years old and according to a manuscript he got out the following year reports he grew old and fell in love. He also mentions a meeting with a woman who broke into his home and first went with him into a bath and gave him food and sleep. Three days later the woman left the room expecting him to eat her lunch and on that day he left home at 9:30 am in despair. He had not been to his bedside. On seeing this, he said a voice in him called out, "Your name is Jack. What is the girl?" Hamilton said the superior told him, "She was a layover in a bed and seven[Pg 209] feet below the bed where the general slept in very feminine attire. Nobody had time to look into her face. What was she to tell you about the general?" A Washington's Official Address to Congress with Americans May 17th, 1781 "I am the one to announce completely that I am a true Christian and an eloquent philosopher. I am not constrained"	1280 bits	Meteor	160-byte payload × 8 bits
"Acclaimed dancer, Luna Moves, unassumingly announced her new song, which intriguingly entwines rhythmic beats with her signature performative flair."	64 bits	LLM-Stega	64 bits/sentence (Subject, Predicate, Object, Emotion)

Stego Text	Cap	Source	Calculation Basis
"The band's praise of their latest song was received with neutrality, a calm before the potential storm of fan reactions."	64 bits	LLM-Stega	64 bits/sentence
"The beloved band joyfully released an immersive exhibition, transporting fans through their rich musical journey, sparking waves of nostalgic happiness across the community."	64 bits	LLM-Stega	64 bits/sentence
"The innovative game developers ecstatically released a gripping new episode, thrilling gamers and inviting them into captivating new worlds."	64 bits	LLM-Stega	64 bits/sentence
"With a voice that touched souls, the singer performed a memorable episode on the live show, casting a warm, positive light across the enchanted audience."	64 bits	LLM-Stega	64 bits/sentence
"I was genuinely underwhelmed by the film, and I'm afraid the only thing that stood out to me was the final scene. I had been eagerly anticipating it, but unfortunately the rest of the movie fell flat for me."	72 bits	Zero-shot	Base64 secret length (12 chars)
"All countries have an ugly side, just as ours does. It's unfair to say that only the Philippines has this issue, though. We are all capable of such shortfalls."	96 bits	Zero-shot	Base64 secret length (16 chars)
"Cheaper doesn't always have to equal inferior, particularly in the case of indie films, where creators are often pushed to maximize the impact of limited resources."	96 bits	Zero-shot	Base64 secret length (16 chars)
"Oh my gawd, stop dting married men! Like seriously, why do we keep falling into this trap over and over? Anglea basset's chracter should totall be more concerned with the fact that her husband abandoned hre for another woma than some mediocore romance flic. Overall, I would give this movie a big fat C- for wasted poasbility and a little more thought."	168 bits	Zero-shot	Base64 secret length (28 chars)
"Yes, I miss this cute kitty very much."	64 bits	GCStego	4 secret words × 16 bits/word
"A good subject to start."	3 bits	ParaLS	Explicit binary watermark "110"
"I was very nervous, at the time. I was 22, and the director was actually at a birthday party in Paris and just saying the doors were closed. But I put on a large black motorcycle, got my album saved and drove on, doing videos about history and the films. So I was really excited. And I found that in this movie, we were at a concert in London. And that concert was where every song and every act was at the movie playing the music that could be heard and the message was, "By God, there's people. But time has passed, and we're here, maybe from another world. Let's see how things can go." And at the end of the movie I said, "I hope the thing happened a bit, because there's no way you can change what you know. You know? Something has happened. Just let it happen. Let it happen." And at the end, I say to myself, "Hey, maybe we can change"	968 bits	Discop	200 tokens × 4.84 bits/token (p = 0.95)

5.3.6 Security Metrics. The sources evaluate security and reliability through the following integrated metrics:

- **Detection Accuracy (Acc) / Error Rate (PE):** Measures a classifier's ability to distinguish between cover and stego text. An accuracy of **50%** (or PE of 50%) is the gold standard, indicating the stego text is statistically identical to normal text.
- **F1 Score:** The harmonic mean of precision and recall, used to verify detection reliability, especially when datasets are balanced.
- **Robustness (Attack Resistance):** Evaluated via the **Mean Impact of Attack (mIOA)** or **Bit Accuracy** after removal attempts such as DAE (Denoising Auto-Encoder), word substitution, or sentence deletion.
- **False Positive Rate (FPR):** Measures how often human or normal model-generated texts are incorrectly flagged as containing secret messages.

Recent robustness-oriented work broadens this evaluation set. Position-agnostic generation tests perturbation tolerance across multiple attack types, active-attack repair methods evaluate message integrity after tampering, and contrastive semantic watermarking reports F1 under rewriting and substitution attacks [6, 24, 42]. This shift suggests that future evaluations should report not only clean-channel capacity and fluency, but also degradation curves under realistic editing.

5.3.7 Evaluation Challenges and Gaps. The broad landscape of evaluation metrics across the reviewed studies is summarized in Table 7, which categorizes metrics by type, lists representative studies, and quantifies their usage frequency. Several significant challenges exist in current evaluation practices:

- **Lack of Standardized Benchmarks:** The absence of shared benchmarks means only 20% of studies use common datasets, making comparison difficult
- **Inconsistent Reporting:** Different units, scales, and methodologies across studies
- **Limited Human Evaluation:** Only 25% of studies include human assessment
- **Missing Robustness Testing:** 60% of studies don't test against various attacks
- **Incomplete Evaluation:** Many studies focus on only one or two metric categories

Table 7. Overview of evaluation metrics and representative studies

Evaluation Metrics	Variant / Category	Representative Studies	Total Count
Imperceptibility	Perplexity (PPL)	VAE-Stega [53], Rewriting-Stego [20], NMT-Stego [8], LLM-Stego [47], Zero-shot [23], DeepStego [18], Emotion-Controllable [37], GCStego [22], RDH [58], ALiSa [54], Semantic Space [56], CPG-LS [49], Principled [15]	13
	Statistical Divergence (KLD, JSD)	VAE-Stega [53], Discop [10], NMT-Stego [8], LLM-Stego [47], Zero-shot [23], Emotion-Controllable [37], PAP [24], Natural [39], Joint-BERT [9], Disambiguating [31]	10
	Semantic Similarity (BLEU, SIM, SS)	NMT-Stego [8], LLM-Stego [47], Emotion-Controllable [37], DeepTextMark [27], Paraphraser-based [32], GCStego [22], Post-hoc [13], Contrastive [42], DeepStego [18], PAP [24], Principled [15], Hi-Stega [41], CWS [19], Beyond Binary [51], Summarization [55], Joint-BERT [9]	13
	Human Evaluation	Emotion-Controllable [37]	1
	Diversity (TTR, Distinct-n)	Multi-criteria [46], Hi-Stega [41], FREmax [29]	3
Security	Detection Accuracy (Acc) / F1-Score	VAE-Stega [53], Rewriting-Stego [20], NMT-Stego [8], LLM-Stego [47], Zero-shot [23], CWS [19], DeepStego [18], Emotion-Controllable [37], PAP [24], GCStego [22], RDH [58], ALiSa [54], Semantic Space [56], Principled [15], DeepTextMark [27], Post-hoc [13], Contrastive [42], Two-Model [6], Natural [39], Semantic Controllable [21], Summarization [55], CPG-LS [49], Meteor [16]	21
Capacity	Bits Per Word / Token (BPW/BPT)	VAE-Stega [53], Rewriting-Stego [20], NMT-Stego [8], LLM-Stego [47], Zero-shot [23], Meteor [16], Principled [15], Hi-Stega [41], Natural [39], Paraphraser-based [32], Semantic Controllable [21], CPG-LS [49], FREmax [29], DeepStego [18], Emotion-Controllable [37], PAP [24], Multi-criteria [46], GCStego [22], RDH [58], Joint-BERT [9], Two-Model [6], Contrastive [42], Post-hoc [13], Beyond Binary [51], Disambiguating [31], ALiSa [54], Semantic Space [56]	27
	Entropy Utilization	Meteor [16], Disambiguating [31], DeepStego [18]	3
Robustness	Recovery Rate / Bit Match Rate	PAP [24], Multi-criteria [46]	2
	Resilience to Attacks (F1 drop)	Post-hoc [13], Principled [15], Beyond Binary [51]	3
Efficiency	Generation / Extraction Time	Meteor [16], Disambiguating [31], DeepStego [18], Contrastive [42], Two-Model [6]	5

5.4 Integration of External Knowledge Sources (RQ4)

The integration of external knowledge sources has emerged as a recurring design pattern in LLM-based steganography. Across the reviewed studies, external information is used primarily to improve contextual plausibility, maintain topic consistency, and in some cases expand the effective space available for embedding.

Table 8 summarizes the main knowledge-integration patterns and their roles.

Table 8. Representative patterns of external knowledge integration

Knowledge Type	Typical Role	Primary Benefit	Examples
Semantic Resources	Topic, entity, and emotion guidance	Better semantic control	Co-Stega [22], knowledge-graph guided methods [9, 21], emotionally controllable steganography [37]
Domain Corpora	Domain adaptation	Better stylistic fit	FREmax [29], specialized datasets
Prompt Engineering	In-context control	Better task alignment	Zero-shot [23], DeepStego [18], multi-criteria optimization [46]
Context Retrieval	Evidence or example retrieval	Higher contextual plausibility	Co-Stega [22], retrieval-augmented pipelines [41]

5.4.1 Semantic Resources Integration. Rather than relying only on the internal prior of the language model, several recent systems inject external guidance during generation. Retrieved posts, headlines, or comments can provide a realistic discourse context for the cover text [41]. Knowledge-graph triples can constrain the content that should appear in the output, improving semantic consistency over longer spans [21]. Named entities and sentiment labels add a finer-grained control layer when the goal is emotional or conversational consistency [37]. Frequency-based external statistics can also be used to steer generation toward distributions that better match the target channel [29]. Taken together, these strategies show that contextual conditioning has become an important mechanism for balancing plausibility, control, and payload.

5.4.2 Domain Corpora Integration. Another recurring pattern is the use of domain-specific corpora. Fine-tuning or conditioning on domain data helps align the generated text with the vocabulary, stylistic conventions, and topical expectations of the intended communication channel. This pattern appears in systems such as VAE-Stega [53], Meteor [16], Hi-Stega [41], FREmax [29], Rewriting-Stego [20], and Summarization-Stego [55]. Although the underlying architectures differ, these methods share the assumption that better alignment with a target domain improves both naturalness and practical usability.

When additional training is not possible, black-box methods move this domain adaptation into the prompt, style specification, or retrieval stage. Zero-shot approaches provide examples directly in the context window, LLM-Stega frames the generation request with topic-constrained prompts, DeepStego uses prompts and synonym guidance, multi-criteria optimization encodes through stylistic controls, and Co-Stega retrieves recent posts to anchor the generated reply in a plausible local context [18, 22, 23, 46, 47]. In these settings, external context functions as a lightweight substitute for model adaptation.

5.4.3 Comprehensive Overview of External Knowledge Integration Patterns. Building on the initial categorization, Table 9 provides a detailed breakdown of how external knowledge sources are integrated across the reviewed studies.

Table 9. Comprehensive overview of external knowledge integration patterns in LLM-based steganography

Approach	**Variant / Category**	**Representative Studies**	**Count**
Structured Knowledge Integration	Knowledge Graphs (KG) & Graph Attention Networks	"A semantic controllable long text steganography framework based on llm prompt engineering and knowledge graph" [21], "Joint linguistic steganography with BERT masked language model and graph attention network" [9]	2
	Information Retrieval (IR): Retrieval-Augmented Generation (RAG) & Corpus Search	"Hi-Stega: A Hierarchical Linguistic Steganography Framework Combining Retrieval and Generation" [41], "Co-Stega: Collaborative Linguistic Steganography for the Low Capacity Challenge in Social Media" [22], "Emotionally Controllable Text Steganography Based on Large Language Model and Named Entity" [37]	3
Linguistic Resource Integration	Paraphrase DBs, Synonym DBs & Dependency Datasets	"Natural language watermarking via paraphraser-based lexical substitution" [32], "DeepStego: Privacy-Preserving Natural Language Steganography Using Large Language Models and Advanced Neural Architectures" [18], "Robust and semantic-faithful post-hoc watermarking of text generated by black-box language models" [13], "Promising Multi-Granularity Linguistic Steganography by Jointing Syntactic and Lexical Manipulations" [28], "Imperceptible Text Steganography based on Group Chat" [19]	5
	External Statistical Analysis: Character/Token Frequency Distributions	"Fremax: A simple method towards truly secure generative linguistic steganography" [29], "Enhancement of the Generation Quality of Generative Linguistic Steganographic Texts by a Character-Based Diffusion Embedding Algorithm (CDEA)" [5]	2
External Model Feedback	Auxiliary Scorer, Critic & Discriminator Models	"Rewriting-Stego: Generating Natural and Controllable Steganographic Text with Pre-trained Language Model" [20], "A Principled Approach to Natural Language Watermarking" [15], "CPG-LS: Causal Perception Guided Linguistic Steganography" [49], "Controllable semantic linguistic steganography via summarization generation" [55], "ALiSa: Acrostic linguistic steganography based on BERT and Gibbs sampling" [54], "DeepTextMark: A Deep Learning-Driven Text Watermarking Approach for Identifying Large Language Model Generated Text" [27], "A Contrastive Semantic Watermarking Framework for Large Language Models" [42]	7
Shared Knowledge / Side Information	Shared Auxiliary Datasets & Public Contexts	"Meteor: Cryptographically secure steganography for realistic distributions" [16], "Zero-shot Generative Linguistic Steganography" [23], "General framework for reversible data hiding in texts based on masked language modeling" [58], "Natural Language Steganography by ChatGPT" [39]	4

5.4.4 Key Insights from Knowledge Integration Patterns. The integration of external knowledge sources follows distinct patterns, each serving specific purposes:

- **Structured Knowledge**: Studies like [21] and [9] use external knowledge graphs or graph attention networks to guide generation with semantic facts, ensuring long-range consistency.
- **Information Retrieval**: Frameworks such as [41] and [22] utilize retrieval modules to search external news datasets or social media platforms for contextually relevant covers.
- **Linguistic Resources**: Works like [32] and [28] leverage external databases (e.g., ParaBank2, PPDB) to ensure synonym substitutions and paraphrases remain semantically faithful.
- **External Model Feedback**: This is the most prevalent category, where models like BERT are used as external discriminators or "soft" scorers to assess the naturalness and security of steganographic output (e.g., [27, 49]).
- **Shared Side Information**: Studies such as [16] and [23] rely on shared (public) models and auxiliary history to synchronize probability distributions between sender and receiver.

5.5 Limitations and Trade-offs in Current Techniques (RQ5)

This subsection consolidates the practical limitations previously introduced in the background discussion and reframes them as findings from the reviewed literature. Across the surveyed studies, the main constraints recur around imperceptibility, effective payload, semantic coherence, tokenization reliability, deployment assumptions, and attack robustness.

5.5.1 The Perceptual-Statistical Imperceptibility Conflict (PSIC Effect). Generative steganography faces an intrinsic tension: optimizing for perceptual fidelity degrades statistical security. Aggressive perplexity (PPL) minimization produces unnaturally “sharp” distributions that deviate from high-variance human writing, increasing detection vulnerability. Higher embedding rates force selection of lower-probability tokens, degrading text quality but expanding the candidate pool to enhance anti-steganalysis resistance. At elevated capacities, output distributions asymptotically approach natural language’s high-entropy chaos, camouflaging signals within legitimate statistical noise [53].

5.5.2 Attack Vulnerability and Security Concerns. Current techniques exhibit a substantial attack surface across multiple vectors, including word-level deletions and lexical substitutions that disrupt dependency parsing. Polishing and re-translation attacks progressively erase watermarks; at high intensities, these structural changes can reduce detection performance to an F1 score of 0.5, rendering the watermark undetectable. Furthermore, adversarial machine learning enables the extraction of secret model parameters from black-box systems, potentially compromising integrity entirely. Recent work responds by moving robustness into the embedding design itself: position-agnostic generation avoids dependence on fixed token locations, active-attack repair models attempt to recover damaged stegotext, and contrastive semantic watermarking tests resilience under rewriting and substitution [6, 24, 42]. To mitigate distributional risks, Variational Auto-Encoders (VAE) are also employed to learn the statistical distribution of normal text, aiming to reduce the KL divergence between cover and stego distributions and resolve the conflict between perceptual and statistical imperceptibility [13].

5.5.3 Capacity and Contextual Constraints. Short, low-entropy contexts severely constrain embedding capacity. Social media posts and technical documents offer minimal redundancy for hiding [22], while semantic preservation requirements compete directly with information embedding objectives. Brief texts further lack sufficient contextual support for effective steganographic operations [8]. Newer approaches attempt to recover some capacity through synonym guidance, stylistic feature control, syntactic-lexical multi-granularity, and reject-split token quality control, but these gains remain tied to the selected prompt, cover text, or candidate-token space [18, 28, 43, 46].

Early generative methods also produced fluent but semantically arbitrary text, violating cognitive imperceptibility [8]. When generation is conditioned only on local token history, the output may drift away from the source topic or conversational logic, which is especially suspicious in low-entropy channels such as social media replies. This contextual drift worsens when methods must keep generating until an arbitrarily large payload is fully embedded: texts become unnaturally long, coherence decays over time, and practical payload efficiency drops even when reported bit capacity appears high [21, 39, 49, 51].

5.5.4 Segmentation and Tokenization Barriers. Subword tokenization introduces critical extraction ambiguities that are **fundamental to white-box steganographic methods**. Byte-pair encoding splits words unpredictably, creating multiple valid segmentations that corrupt message boundaries and cause decoding failures. These ambiguities impose hard capacity caps regardless of algorithmic sophistication—**SparSamp** suffers accuracy degradation from token ambiguity (TA), while **ShiMer** cannot effectively boost entropy due to tokenization constraints [31].

The core issue lies in **segmentation and tokenization barriers**-also termed **segmentation ambiguity**-which persist **even when sender and receiver employ identical Large Language Models and tokenizers**. Because modern LLMs decompose words into subword units, and vocabularies frequently contain both complete words and their constituent components, any given text string admits multiple valid token representations. This multiplicity is compounded by the **detokenization-retokenization gap**: senders must convert tokens to plain text to conceal their transmission, while receivers must subsequently retokenize this text to recover hidden messages. Despite shared tokenization rules, receivers may select different token sequences than senders intended-senders might encode messages using separate tokens (" any" and "thing"), only for receivers to retokenize "anything" as a **single token** (" anything"). Because LLMs operate **autoregressively**, a single divergent token choice alters the probability distribution for all subsequent predictions, causing **total decoding failure** for the remainder of the message. To address this **white-box-specific challenge**, researchers propose **SyncPool**, a mechanism that aggregates ambiguous tokens into pools and employs synchronized random number generators to ensure both parties select **identical tokens**, thereby eliminating ambiguity and guaranteeing reliable message extraction. [31]

5.5.5 White-box versus Black-box Trade-offs. The architectural choice between white-box and black-box access paradigms involves fundamental trade-offs. Across our reviewed corpus of 32 studies, only **2** papers adopt black-box approaches, while the remaining 30 employ white-box or hybrid methods. White-box methods typically assume access to internal language-model components such as token probabilities, decoding states, vocabularies, or model parameters, and some approaches additionally require expensive fine-tuning or retraining procedures to align the model distribution with steganographic objectives. While these methods can achieve strong statistical imperceptibility through distribution-aware generation (e.g., VAE-Stega and arithmetic-coding-based methods), they suffer from reduced accessibility, high computational cost, and deployment complexity.

In contrast, black-box approaches operate through standard LLM APIs or user interfaces without requiring direct access to model internals or retraining, substantially improving practicality and usability. However, the distinction is primarily about *model access*, not the absence of probabilistic coordination, as several black-box approaches still rely on shared probabilistic mechanisms, semantic distributions, or synchronized auxiliary structures between sender and receiver. Recent semantic black-box methods also prioritize robustness against paraphrasing, token perturbation, and noisy communication channels rather than purely token-level imperceptibility. Among the few peer-reviewed black-box methods currently available, **LLM-Stega** reports a capacity of 5.93 bpw with perplexity 165.76, illustrating that improved accessibility does not eliminate the PSIC trade-off between payload capacity, text quality, robustness, and security. Several recent systems further blur strict categorical boundaries by combining retrieval-, generation-, editing-, or substitution-based paradigms within hybrid steganographic frameworks.

5.5.6 Computational and Resource Constraints. Performance optimization conflicts directly with computational efficiency [16]. Superior results demand increased resources and memory, particularly for large models with external knowledge bases [10]. Real-time latency constraints limit optimization depth, while performance degradation emerges at operational scale. Variable length sampling, rejection sampling, active repair, and token-level quality control significantly increase encoding or decoding complexity, and LLM-based steganography requires 3x to 5x more time than prior methods [31, 43]. The **Meteor** system exemplifies hardware constraints, running nearly 50× slower on mobile devices compared to GPU environments.

LLM-specific deployment obstacles compound these costs. Hallucinations and nonsensical continuations can corrupt the covert bitstream or create detectable artifacts, while lossy transmission, editing, or platform-side normalization can irreversibly damage extraction. As model quality improves, steganalysis systems also improve, producing an arms race in which better generation does not automatically translate into better security.

5.5.7 Critical Unresolved Challenges. Foundational gaps persist across the research landscape: theoretical security guarantees remain unestablished, resilience to advanced attacks is limited, evaluation frameworks lack standardization, responsible development guidelines are absent, non-English performance lags substantially, and real-world deployment testing remains insufficient. Several studies also surface unresolved concerns around reversibility, bias, and misuse potential, indicating that robust engineering alone will not be enough without clearer security and governance models [1, 23, 32, 58].

5.5.8 Quantified Impact Summary. Quantitative evidence highlights several constraints on current steganographic systems. The Perceptual-Statistical Imperceptibility Conflict (PSIC) effect creates a trade-off where increasing the embedding rate (e.g., to 5 bpw) strengthens anti-steganalysis but significantly degrades perceptual text quality [53]. Attack vulnerability is notable; for example, adaptive attacks can reduce bit accuracy to near-random levels (0.6) [15]. Segmentation ambiguity and entropy variability in communication channels are major impediments that lead to decoding failures and restricted practical capacity [16]. Finally, computational optimizations induce significant overhead, with some reordering algorithms causing up to 69% performance degradation in processing speed [10].

6 DISCUSSION

This section synthesizes the findings across the five research questions and discusses how recent LLM-based approaches are changing the design priorities of linguistic steganography.

6.1 The Emergence of Meaning-Based Steganography

One of the clearest patterns in the reviewed studies is the shift from strictly white-box, probability-driven methods toward broader black-box, context-aware, and robustness-oriented approaches. Earlier linguistic steganography often depended on local lexical or sampling manipulations that were difficult to scale and easy to distort semantically. More recent work uses LLMs to support full-text generation, rewriting, prompting, retrieval, stylistic control, and token-level quality control, thereby moving the problem from isolated token selection toward broader control of discourse, context, and transmission resilience.

6.2 Resolving the PSIC Effect through Contextual Compatibility

The Perceptual-Statistical Imperceptibility Conflict (PSIC) remains one of the main technical hurdles identified across the reviewed studies. The synthesis suggests that **contextual compatibility** is not simply an optional enhancement, but an increasingly important strategy for mitigating this conflict.

By integrating external knowledge sources (RQ4), recent systems can better align generated text with the topic, style, and discourse expectations of a target channel. In practical terms, this means that security is increasingly tied not only to distributional similarity, but also to whether the text remains semantically plausible within a recognizable communicative setting.

6.3 Implications for Research and Practice

6.3.1 The Democratization of Covertness. The shift toward black-box approaches (RQ1, RQ2) lowers the practical barrier to deployment. By treating models as opaque APIs or hosted services, covert communication is no longer limited to settings where both parties can share and run the same model locally. This may improve accessibility in constrained environments, but as noted in RQ5 it also lowers the barrier for misuse.

6.3.2 The Need for Semantic-First Evaluation. The evaluation metrics reviewed in RQ3 show that the field still relies heavily on surface-level or model-dependent metrics such as perplexity and bit-density measures, while human and semantic evaluation remain less common. The new 2025 studies add stronger robustness and capacity claims, but they also reinforce the same comparability problem because reported improvements depend on different baselines, datasets, prompt settings, and attack models. Future benchmarking should therefore place more emphasis on semantic and human-centered evaluation alongside statistical measures and realistic perturbation tests.

6.4 Ethical Responsibility and the Forensics Arms Race

The reviewed literature also highlights a gap between technical innovation and explicit discussion of misuse, governance, and defensive implications. As LLM-based steganography becomes more fluent and easier to deploy, steganalysis and provenance research will likely need to adapt beyond traditional lexical or shallow statistical features.

6.5 Limitations of the Review

This review should be interpreted as a structured snapshot of a rapidly changing field. The pace of LLM development, the heterogeneity of evaluation settings, and the exclusion of six records pending PDF acquisition all limit the stability of any quantitative summary.

6.6 Future Research Directions

Based on the identified gaps, four research directions appear especially important:

- **Provable Semantic Security**: Extending security analysis beyond token distributions toward stronger guarantees about contextual plausibility and recoverability.
- **Robustness-Aware Capacity**: Reporting payload together with degradation under paraphrasing, substitution, deletion, platform normalization, and active tampering.

- **Multimodal Grounding**: Studying how text-based hiding strategies interact with multimodal systems without collapsing the review boundary between steganography and adjacent security problems.
- **Adaptive Context Synthesis**: Designing retrieval and prompting pipelines that improve realism without making systems excessively fragile, slow, or difficult to evaluate.

7 THREATS TO VALIDITY

Several threats to validity should be considered when interpreting the findings of this review. First, the search was limited to English-language studies published from 2018 onward. This boundary is consistent with the LLM focus of the review, but it may exclude earlier linguistic steganography work or relevant non-English publications.

Second, although the selected databases provide broad coverage, the search query may have missed studies that use different terminology for related techniques, especially work framed as covert communication, text watermarking, or information hiding without explicitly mentioning LLMs. Snowballing was used to reduce this risk, but it cannot guarantee complete coverage.

Third, the review relies on the information reported in the primary studies. Because evaluation settings, datasets, baselines, and attack models differ substantially across papers, the reported results are not always directly comparable. This limits the strength of quantitative conclusions and makes the synthesis primarily descriptive.

Fourth, the selection and extraction process may include researcher judgment, especially when classifying application domains, context-awareness mechanisms, and reported limitations. To mitigate this threat, the extraction form used predefined categories aligned with the research questions, and the results were summarized consistently across studies.

Finally, six potentially relevant papers were not included in the final synthesis because full PDFs or complete extraction data were not available at the time of analysis. Their exclusion may slightly affect the reported proportions, but the main qualitative trends are unlikely to change substantially.

8 CONCLUSION

This systematic literature review shows that Large Language Models (LLMs) have substantially expanded the design space of linguistic steganography. Across 34 primary studies, the reviewed literature points to a shift toward generative, rewriting-based, black-box, context-aware, and robustness-oriented techniques that aim to improve fluency and contextual plausibility while maintaining usable payload.

At the same time, the review shows that these advances do not remove the core constraints of the field. Evaluation remains inconsistent, comparisons across studies are often difficult, and persistent trade-offs remain among imperceptibility, capacity, robustness, and deployability. In particular, the Perceptual-Statistical Imperceptibility Conflict (PSIC), low-entropy communication settings, tokenization ambiguity, and black-box limitations remain central obstacles.

One of the clearest themes across the reviewed studies is the growing importance of contextual compatibility. Domain adaptation, retrieval, semantic guidance, entity and emotion control, stylistic constraints, and token-level quality control are increasingly used to make stego text more plausible within a target communicative setting. This suggests that future progress will depend not only on better language models, but also on stronger evaluation protocols and more disciplined integration of context.

Future work should therefore focus on standardized evaluation, more reliable extraction under realistic tokenization and editing conditions, robustness-aware capacity reporting, stronger security guarantees for practical systems, and clearer boundaries between linguistic steganography and adjacent areas such as watermarking or broader AI security.

Overall, the literature suggests that LLMs have made linguistic steganography more practical and expressive, but not yet methodologically mature.

REFERENCES

[1] Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big?. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*. ACM, Virtual Event, Canada, 610–623.

[2] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. 2020. Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems* 33 (2020), 1877–1901.

[3] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, et al. 2023. Sparks of Artificial General Intelligence: Early Experiments with GPT-4. *arXiv preprint arXiv:2303.12712* (2023).

[4] Christian Cachin. 1998. An Information-Theoretic Model for Steganography. In *Information Hiding*, David Aucsmith (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 306–318.

[5] Yingquan Chen, Qianmu Li, Aniruddha Bhattacharjya, Xiaocong Wu, Huifeng Li, Qing Chang, Le Zhu, and Yan Xiao. 2025. Enhancement of the Generation Quality of Generative Linguistic Steganographic Texts by a Character-Based Diffusion Embedding Algorithm (CDEA). *Applied Sciences* 15, 17 (2025), 9663. doi:10.3390/app15179663

[6] Yingquan Chen, Qianmu Li, Xiaocong Wu, and Zijian Ying. 2025. Research on Making Two Models Based on the Generative Linguistic Steganography for Securing Linguistic Steganographic Texts from Active Attacks. *Symmetry* 17, 9 (2025), 1416. doi:10.3390/sym17091416

[7] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 4171–4186.

[8] Changhao Ding, Zhangjie Fu, Zhongliang Yang, Qi Yu, Daqiu Li, and Yongfeng Huang. 2023. Context-aware linguistic steganography model based on neural machine translation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 32 (2023), 868–878.

[9] Changhao Ding, Zhangjie Fu, Qi Yu, Fan Wang, and Xianyi Chen. 2023. Joint linguistic steganography with BERT masked language model and graph attention network. *IEEE Transactions on Cognitive and Developmental Systems* 16, 2 (2023), 772–781.

[10] Jinyang Ding, Kejiang Chen, Yaofei Wang, Na Zhao, Weiming Zhang, and Nenghai Yu. 2023. Discop: Provably secure steganography in practice based on distribution copies. In *2023 IEEE Symposium on Security and Privacy (SP)*. IEEE, San Francisco, CA, USA, 2238–2255.

[11] Lizhe Fang, Yifei Wang, Zhaoyang Liu, Chenheng Zhang, Stefanie Jegelka, Jinyang Gao, Bolin Ding, and Yisen Wang. 2025. What is wrong with perplexity for long-context language modeling?. In *International Conference on Learning Representations*, Vol. 2025. 94541–94563.

[12] Jessica Fridrich. 2009. *Steganography in digital media: principles, algorithms, and applications*. Cambridge University Press, Cambridge, UK.

[13] Jifei Hao, Jipeng Qiang, Yi Zhu, Yun Li, Yunhao Yuan, Xiaocheng Hu, and Xiaoye Ouyang. 2025. Robust and semantic-faithful post-hoc watermarking of text generated by black-box language models. *Frontiers of Computer Science* 19, 9 (2025), 199357.

[14] Fred Jelinek, Robert L Mercer, Lalit R Bahl, and James K Baker. 1977. Perplexity—a measure of the difficulty of speech recognition tasks. *The journal of the Acoustical Society of America* 62, S1 (1977), S63–S63.

[15] Zhe Ji, Qiansiqi Hu, Yicheng Zheng, Liyao Xiang, and Xinbing Wang. 2024. A principled approach to natural language watermarking. In *Proceedings of the 32nd ACM International Conference on Multimedia*. ACM, 2908–2916.

[16] Gabriel Kaptchuk, Tushar M Jois, Matthew Green, and Aviel D Rubin. 2021. Meteor: Cryptographically secure steganography for realistic distributions. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*. ACM, Virtual Event, Republic of Korea, 1529–1548.

[17] S. Kullback and R. A. Leibler. 1951. On Information and Sufficiency. *The Annals of Mathematical Statistics* 22, 1 (1951), 79–86. http://www.jstor.org/stable/2236703

[18] Oleksandr Kuznetsov, Kyrylo Chernov, Aigul Shaikhanova, Kainizhamal Iklassova, and Dinara Kozhakhmetova. 2025. DeepStego: Privacy-Preserving Natural Language Steganography Using Large Language Models and Advanced Neural Architectures. *Computers* 14, 5 (2025), 165. doi:10.3390/computers14050165

[19] Fanxiao Li, Ping Wei, Tingchao Fu, Yu Lin, and Wei Zhou. 2024. Imperceptible Text Steganography based on Group Chat. In *2024 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 1–6.

[20] Fanxiao Li, Sixing Wu, Jiong Yu, Shuoxin Wang, BingBing Song, Renyang Liu, Haoseng Lai, and Wei Zhou. 2023. Rewriting-Stego: generating natural and controllable steganographic text with pre-trained language model. In *International Conference on Database Systems for Advanced Applications*. Springer, 617–626.

[21] Yihao Li, Ru Zhang, Jianyi Liu, and Qi Lei. 2024. A semantic controllable long text steganography framework based on llm prompt engineering and knowledge graph. *IEEE Signal Processing Letters* (2024).

[22] Guorui Liao, Jinshuai Yang, Kaiyi Pang, and Yongfeng Huang. 2024. Co-stega: Collaborative linguistic steganography for the low capacity challenge in social media. In *Proceedings of the 2024 ACM Workshop on Information Hiding and Multimedia Security*. ACM, Baiona, Spain, 7–12.

[23] Ke Lin, Yiyang Luo, Zijian Zhang, and Luo Ping. 2024. Zero-shot generative linguistic steganography. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 5168–5182.

[24] Shuo Lin, Zhenyang Shen, Xiang Li, Jiacheng Fan, and Sixing Wu. 2025. Position-Agnostic Probabilistic Generation for Robust Steganographic Text. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*. ACM, 4950–4954. doi:10.1145/3746252.3760861

[25] Mohammed Abdul Majeed, Rossilawati Sulaiman, Zarina Shukur, and Mohammad Kamrul Hasan. 2021. A Review on Text Steganography Techniques. *Mathematics* 9, 21 (2021). doi:10.3390/math9212829

[26] Abby Morgan. 2024. Perplexity for LLM Evaluation. Comet ML Blog. https://www.comet.com/site/blog/perplexity-for-llm-evaluation/ Accessed: 2025-12-31.

[27] Travis Munyer, Abdullah All Tanvir, Arjon Das, and Xin Zhong. 2024. DeepTextMark: a deep learning-driven text watermarking approach for identifying large language model generated text. *Ieee Access* 12 (2024), 40508–40520.

[28] Chengfu Ou, Lingyun Xiang, and Yangfan Liu. 2025. Promising Multi-Granularity Linguistic Steganography by Jointing Syntactic and Lexical Manipulations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. AAAI Press, 24984–24992. doi:10.1609/aaai.v39i23.34682

[29] Kaiyi Pang, Minhao Bai, Jinshuai Yang, Huili Wang, Minghu Jiang, and Yongfeng Huang. 2024. Fremax: A simple method towards truly secure generative linguistic steganography. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 4755–4759.

[30] Kai Petersen, Sairam Vakkalanka, and Ludwik Kuzniarz. 2015. Guidelines for conducting systematic mapping studies in software engineering: An update. *Information and Software Technology* 64 (08 2015). doi:10.1016/j.infsof.2015.03.007

[31] Yuang Qi, Kejiang Chen, Kai Zeng, Weiming Zhang, and Nenghai Yu. 2024. Provably secure disambiguating neural linguistic steganography. *IEEE Transactions on Dependable and Secure Computing* (2024). Early Access.

[32] Jipeng Qiang, Shiyu Zhu, Yun Li, Yi Zhu, Yunhao Yuan, and Xindong Wu. 2023. Natural language watermarking via paraphraser-based lexical substitution. *Artificial Intelligence* 317 (2023), 103859.

[33] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. *Language Models are Unsupervised Multitask Learners*. Technical Report. OpenAI.

[34] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language Models are Unsupervised Multitask Learners. *OpenAI Blog* (2019). https://d4mucfpksywv.cloudfront.net/better-language-models/language_models_are_unsupervised_multitask_learners.pdf

[35] Emily Reif, Daphne Ippolito, Ann Yuan, Andy Coenen, Chris Callison-Burch, and Jason Wei. 2022. A recipe for arbitrary text style transfer with large language models. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. 837–848.

[36] De Rosal Ignatius Moses Setiadi, Sudipta Kr Ghosal, and Aditya Kumar Sahu. 2025. AI-Powered Steganography: Advances in Image, Linguistic, and 3D Mesh Data Hiding – A Survey. *Journal of Future Artificial Intelligence and Technologies* 2, 1 (Apr. 2025), 1–23. doi:10.62411/faith.3048-3719-76

[37] Hao Shi, Wenpu Guo, and Shaoyuan Gao. 2025. Emotionally Controllable Text Steganography Based on Large Language Model and Named Entity. *Technologies* 13, 7 (2025), 264. doi:10.3390/technologies13070264

[38] Gustavus J Simmons. 1984. The prisoners' problem and the subliminal channel. In *Advances in Cryptology: Proceedings of Crypto 83*. Springer, Boston, MA, 51–67.

[39] Martin Steinebach. 2024. Natural language steganography by chatgpt. In *Proceedings of the 19th International Conference on Availability, Reliability and Security*. ACM, 1–9.

[40] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).

[41] Huili Wang, Zhongliang Yang, Jinshuai Yang, Yue Gao, and Yongfeng Huang. 2023. Hi-stega: A hierarchical linguistic steganography framework combining retrieval and generation. In *International Conference on Neural Information Processing*. Springer, 41–54.

[42] Jianxin Wang, Xiangze Chang, Chaoen Xiao, and Lei Zhang. 2025. A Contrastive Semantic Watermarking Framework for Large Language Models. *Symmetry* 17, 7 (2025), 1124. doi:10.3390/sym17071124

[43] Pengtao Wang, Jianfeng Qu, Yu Liu, and Yanmei Kang. 2025. RS-Stega: A Token-Level Quality-Control Framework for Generative Text Steganography. In *Web and Big Data (Lecture Notes in Computer Science, Vol. 16114)*. Springer, 80–97. doi:10.1007/978-981-95-5716-5_6

[44] Yequan Wang, Jiawen Deng, Aixin Sun, and Xuying Meng. 2022. Perplexity from plm is unreliable for evaluating text quality. *arXiv preprint arXiv:2210.05892* (2022).

[45] Yihao Wang, Ru Zhang, Yifan Tang, and Jianyi Liu. 2023. State-of-the-art Advances of Deep-learning Linguistic Steganalysis Research. In *2023 International Conference on Data, Information and Computing Science (CDICS)*. IEEE, 20–24. doi:10.1109/cdics61497.2023.00014

[46] Kamil Woźniak, Marek R. Ogiela, and Lidia Ogiela. 2025. Multi-criteria linguistic optimization for covert communication in secure LLM-based steganography. *Applied Soft Computing* 185 (2025), 113960. doi:10.1016/j.asoc.2025.113960

[47] Jiaxuan Wu, Zhengxian Wu, Yiming Xue, Juan Wen, and Wanli Peng. 2024. Generative text steganography with large language model. In *Proceedings of the 32nd ACM International Conference on Multimedia*. ACM, Melbourne, Australia, 10345–10353.

[48] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. 2025. The rise and potential of large language model based agents: A survey. *Science China Information Sciences* 68, 2 (2025), 121101.

[49] Lingyun Xiang, Jiali Xia, Yangfan Liu, and Yan Gui. 2023. CPG-LS: Causal Perception Guided Linguistic Steganography. *IEEE Signal Processing Letters* 30 (2023), 1762–1766.

[50] Jianfei Xiao, Yancan Chen, Yimin Ou, Hanyi Yu, Kai Shu, and Yiyong Xiao. 2024. Baichuan2-sum: Instruction finetune baichuan2-7b model for dialogue summarization. In *2024 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 1–8.

[51] Zhenyu Xu, Ruoyu Xu, and Victor S Sheng. 2024. Beyond binary classification: Customizable text watermark on large language models. In *2024 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 1–8.

[52] Aiyuan Yang, Bin Xiao, Binyuan Wang, Binxin Zhang, Ce Bian, Chao Yin, Chenxu Lv, Da Pan, Dian Wang, Dong Yan, et al. 2023. Baichuan 2: Open Large-scale Language Models. *arXiv preprint arXiv:2309.10305* (2023).

[53] Zhong-Liang Yang, Si-Yu Zhang, Yu-Ting Hu, Zhi-Wen Hu, and Yong-Feng Huang. 2020. VAE-Stega: linguistic steganography based on variational auto-encoder. *IEEE Transactions on Information Forensics and Security* 16 (2020), 880–895.

[54] Biao Yi, Hanzhou Wu, Guorui Feng, and Xinpeng Zhang. 2022. ALiSa: Acrostic linguistic steganography based on BERT and Gibbs sampling. *IEEE Signal Processing Letters* 29 (2022), 687–691.

[55] Ruifan Zhang, Jianyi Liu, and Ru Zhang. 2024. Controllable semantic linguistic steganography via summarization generation. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 4560–4564.

[56] Siyu Zhang, Zhongliang Yang, Jinshuai Yang, and Yongfeng Huang. 2020. Linguistic steganography: From symbolic space to semantic space. *IEEE Signal Processing Letters* 28 (2020), 11–15.

[57] Siyu Zhang, Zhongliang Yang, Jinshuai Yang, and Yongfeng Huang. 2021. Provably secure generative linguistic steganography. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*. 3046–3055.

[58] Xiaoyan Zheng, Yurun Fang, and Hanzhou Wu. 2022. General framework for reversible data hiding in texts based on masked language modeling. In *2022 IEEE 24th International Workshop on Multimedia Signal Processing (MMSP)*. IEEE, 1–6.

Table 10. Summary of Results from Reviewed Papers

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
VAE-Stega: linguistic steganography based on va...	BERTBASE (BERT-LSTM) (LSTM-LSTM) model was trained from scratch	2020.0	Twitter (2.6M sentences) IMDB (1.2M sentences) preprocessed	PPL: 28.879, ΔMP: 0.242, KLD: 3.302, JSD: 10.411, Acc: 0.600, R: 0.616	non-explicit	pre-text	text
General framework for reversible data hiding in...	BERTBase	2022.0	BookCorpus	BPW=0.5335 F1=0.9402 PPL=134.2199	non-explicit	pre-text	text
Co-stega: Collaborative linguistic steganograph...	Llama-2-7B-chat, GPT-2 (fine-tuned), Llama-2-13B	2024.0	Tweet dataset (for GPT-2 fine-tuning), Twitter (real-time testing)	SR1: 60.87%, SR2: 98.55%, Gen. Capacity: 44.91 bits, Entropy: 49.21 bits, BPW: 2.31, PPL: 16.75, SimCSE: 0.69	explicit	Social Media	text

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Joint linguistic steganography with BERT masked...	LSTM + attention for temporal context. GAT for spatial token relationships. BERT MLM for deep semantic context in substitution.	2023.0	OPUS	PPL=13.917 KLD=2.904 SIM=0.812 ER=0.365 (BN=2) Best Acc=0.575 (BERT classifier) FLOPs=1.834G	explicit	pre-text	text
Discop: Provably secure steganography in practi...	GPT-2	2023.0	IMDB	p=1.00 Total Time (seconds)=362.63 Ave Time ↓ (seconds/bit)=6.29E-03 Ave KLD ↓ (bits/token)=0 Max KLD :.	non-explicit	tuning + pretext	text

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Generative text steganography with large langua...	Any	2024.0	[Not specified]	Length: 13.333 (words). BPW: 5.93 bpw PPL: 165.76. Semantic Similarity (SS): 0.5881 LS-CNN Acc: 51.55%. BiLSTM-Dense Acc: 49.20%. Bert-FT Acc: 50...	explicit	[Not specified]	[Not specified]
Meteor: Cryptographically secure steganography ...	GPT-2	2021.0	Hutter Prize, HTTP GET requests	GPT-2: 3.09 bits/token	non-explicit	tuning + pretext	text
Zero-shot generative linguistic steganography	LLaMA2-Chat-7B (as the stegotext generator / QA model). GPT-2 (for NLS baseline and JSD evaluation)	2024.0	IMDB, Twitter	PPL: 8.81. JSDfull: 17.90 (x10[truncated]iicircum-2). JSDhalf: 16.86 (x10[truncated]iicircum-2). JSDzero: 13.40 (x10[truncated]iicircum-2) TS...	explicit	zero-shot + prompt	text

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Provably secure disambiguating neural linguisti...	LLaMA2-7b (English), Baichuan2-7b (Chinese)	2024.0	IMDb dataset (100 texts/sample, 3 English sentences + Chinese translations)	Total Error: 0%, Ave KLD: 0, Max KLD: 0, Ave PPL: 3.19 (EN), 7.49 (ZH), Capacity: 1.03–3.05 bits/token, Utilization: 0.66–0.74, Ave Time: [trunc...	non-explicit	pretext	text
A principled approach to natural language water...	Transformer-based encoder/decoder; BERT for distillation	2024.0	Web Transformer 2	Bit acc: 0.994 (K=None), 1.000 (DAE), 0.978 (Adaptive+K=S); Meteor Drop: [truncated]iitilde0.057; SBERT ↑: [truncated]iitilde...	Yes; semantic-level embedding; synonym substitution using BERT	Yes; watermark message assigned categorical label (e.g., 4-bit → 1-of-16)	Yes; semantic embeddings via transformer encoder and BERT; SBERT distance as metric
Context-aware linguistic steganography model ba...	BERT (encoder), LSTM (decoder)	2024.0	WMT18 News Commentary (train/test), Yang et al. bits, Doc2Vec, 5,000 stego pairs (8:1:1 split)	BLEU: 30.5, PPL: 22.5, ER: 0.29, KL: 0.02, SIM: 0.86, Stego detection [truncated]iitilde16%	Yes	[Not specified]	GCF (global context), LMR (language model reference), Multi-head attention

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
DeepTextMark: a deep learning-driven text water...	Model-independent; tested with OPT-2.7B	2024.0	Dolly ChatGPT (train/validate), C4 (test), robustness & sentence-level test sets	100% accuracy (multi-synonym, 10-sentence), mSMS: 0.9892, TPR: 0.83, FNR: 0.17, Detection: 0.00188s, Insertion: 0.27931s	NO	[Not specified]	[Not specified]
Hi-stega: A hierarchical linguistic steganograp...	GPT-2	2024.0	Yahoo! News (titles, bodies, comments); 2,400 titles used	ppl: 109.60, MAUVE: 0.2051, ER2: 10.42, Δ(cosine): 0.0088, Δ(simcse): 0.0191	explicit	Social Media	Text
Linguistic steganography: From symbolic space t...	CTRL (generation), BERT (semantic classifier)	2020.0	5,000 CTRL-generated texts per semanteme (n = 2–16); 1,000 user-generated texts for anti-steganalysis	Classifier Accuracy: 0.9880; Loop Count: 1.0160; PPL: 13.9565; Anti-Steganalysis Accuracy: [truncated]iitilde0.5	implicit	Text	Semanteme (α) as a vector in semantic spac

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Natural language steganography by chatgpt	[Not specified]	2024.0	Custom word sets for specific topics (e.g., 16×10-word sets for music reviews)	[Not specified]	Explicit	Specific Genre/Topic Text	Text
Natural language watermarking via paraphraser-b...	Transformer (Paraphraser), BART (BARTScore), BERT (BLEURT, comparisons)	2023.0	ParaBank2, LS07, CoInCo, Novels, WikiText-2, IMDB, NgNews	LS07 P@1: 58.3, GAP: 65.1; CoInCo P@1: 62.6, GAP: 60.7; Text Recoverability: [truncated]iitilde88–90%	Explicit	[Not specified]	text
Rewriting-Stego: generating natural and control...	BART (bart-base2)	2023.0	Movie, News, Tweet	BPTS: 4.0, BPTC+S: 4.0, PPL: 62.1, Mean: 44.4, Variance: 2.1e04, Acc: 8.9%	not Explicit	[Not specified]	[Not specified]

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
ALiSa: Acrostic linguistic steganography based ...	BERT (Google's BERTBase, Uncased)	2022.0	BookCorpus (10,000 natural texts for evaluation)	PPL: Natural = 13.91, ALiSa = 14.85; LS-RNN/LS-BERT Acc & F1 = [truncated]iitilde0.50; Outperforms GPT-AC/ADG in all cases	No	[Not specified]	[Not specified]
Imperceptible Text Steganography based on Group...	Qwen-7B-Chat	2024.0	HC3, DailyDialogue, COCO Descriptions	HC3: Bit 188.94, Stego 131.99, PPL 34.07, Mean 20.19, Var 0.1e04, F1 90.01%; DailyDialogue: Bit 188.94, Stego 89.37, PPL 53.88, Mean 20.13, Var 0....	Explicit	Social Media / Group Chat	Text (chat history and current input)
A Semantic Controllable Long Text Steganography...	Llama 7B Chat, Meta LLaMA2 7B Chat	2024.0	Story (ChatGPT), Post (Recipe Kaggle + ChatGPT), Ad (Mobile Kaggle + ChatGPT)	ppl ↓ >23%, Δppl ↓ >72% vs ADG/HC/Bin; detection accuracy ↓ >...	Explicit	Topical Content	KG triplets (e1, r, e2), task descriptions (D)

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Beyond Binary Classification: Customizable Text...	gpt-3.5-turbo-instruct, OPT-6.7b, babbage-002, davinci-002 (others: ChatGPT, GPT-2–4, LLaMA)	2024.0	Realnewslike (C4, 500 samples, 100-token prompts + completions); Custom watermark dataset (short info <10 tokens)	AUC 0.98, FPR 0.00, FNR 0.00, [truncated]iitilde100% single-letter decoding, PPL close to human text	Implicit	General Text Generation	Text (evolving prompt + generated output)
CPG-LS: Causal Perception Guided Linguistic Ste...	BERTBase, Cased	2023.0	CC-100 corpus; 10k cover texts; 7:3 train-test split	PPL 36.5; Mauve 0.871; Payload 0.150 bits/word; BiLSTM-D Acc 0.387 F1 0.375; R-BI-C Acc 0.378 F1 0.366; TS-RNN Acc 0.380 F1 0.368	Implicit	Natural Language Text	Text, embeddings, vector matrix

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Controllable Semantic Linguistic Steganography ...	BERT + CRF	2024.0	Gigaword; CNN/Daily Mail	Rouge-1: 0.2212; Rouge-2: 0.0268; Rouge-L: 0.1609; Meteor: 0.1384; Cosine: 0.5911; Euclidean: 5.6386; Manhattan: 87.9534; Jaccard: 0.2022; Anti-ste...	Explicit	Social Media	Semantic features of input text; 384-dim dense vectors for evaluation
FREmax: A Simple Method Towards Truly Secure Ge...	GPT-2	2024.0	Tweet corpus (2.6M sents, 26.8M tokens), IMDB corpus (1.05M sents, 25.3M tokens)	Tweet: PPL 361.83, Entropy 48.21, Tokens 10.83, Distinct3 0.98, BPS 62.79, SI% 73.03. IMDB: PPL 169.66, Entropy 103.39, Tokens 23.80, Distinct3 0....	Implicit	General Text	N-gram frequency distribution stored in a look-up table

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Robust and semantic-faithful post-hoc watermark...	Any	2025.0	HC3 corpus (wiki_csai, open_qa, medicine, reddit_eli5); 200 ChatGPT texts per dataset ([truncated]iitilde200 tokens); half watermarked	FPR: 0.05, TNR: 0.95, TPR: 1.0, FNR: 0.00, Acc: 0.97, Sim: 0.9956±.005, F1≈0.5 under strong doc attacks	Implicit	Informational Text	Text, Pretext (preceding word), Graph (dependency tree)
Contrastive Semantic Watermarking Framework	GPT-2, OPT-1.3B, LLaMA-7B	2025.0	C4 and DBpedia Class benchmark.	C4/DBpedia (GPT-2 Beam): Acc 99.4%, F1 99.9%, Stealth PPL 7.3. Robustness (Semantic Rewriting): F1 93.7%, BLEU-4 78.5, ROUGE-L 0.79.	Implicit	General Text	Vector (shared embeddings)

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
DeepStego	GPT-4-omni	2025.0	Custom dataset of 8,662 samples (generated by GPT-4-omni).	Enhanced Mode (128 bits): Capacity 16.64%, BPW 9.84, Readability 23.40, Natural Language 0.863, Semantic Coherence 0.912. Steganalysis Resistance:...	Implicit	General Text	Text/Vector (embedding similarity)
Emotionally Controllable Steganography	LLaMA-3, Qwen-2.5, GPT-2	2025.0	IMDB movie review dataset (33,000 preprocessed samples).	IMDB (LLaMA-3 3B + Proposed): ER 2.38 bit/word, PPL 16.30, KLD 17.95, BLEU 68.86. Anti-steganalysis (BERT): Acc 0.738, Recall 0.739.	Explicit	Social Media	Text/Label (entities + sentiment)

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Multi-criteria Linguistic Optimization	GPT-4o, Claude-3.7, etc.	2025.0	Six generation contexts (e.g., travel diary, financial report, treasure map narrative).	Aggregated (k=5): BPT 0.24-0.28, PPL 36.92, TTR 0.59, FRE 44.32, GRU 0.54. Aggregated (k=6): BPT 0.26-0.37, PPL 41.20, TTR 0.62, FRE 43.63, GRU 0.58.	Implicit	General Text	Text/Vector (sentence feature window)
Position-Agnostic Generation	GPT-2	2025.0	Large Movie Review Dataset (100,000 IMDB reviews).	PAP (rwc=3): PPL 8.57, BPW 0.17, KLD 2.21. Robustness (Bit Match Rate %): First Word 100, Delete 100, Substitute 100, Paraphrase 96.62, Noise 100.	Implicit	General Text	Vector (embeddings + clusters)

Continued on next page

Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Two-Model Defense & Repair	GPT-2	2025.0	6,000 GPT-2 generated texts; evaluated in Wi-Fi, Bluetooth, IoMT, and IIoT environments.	ISSA-D (CP=32): KL 7.23, ppl 63.99, Div 0.73, Des 0.41, bpw 0.41, FTR 100%. DODRTA-D (CP=2): KL 4.70, ppl 29.0, Div 0.84, bpw 0.96, FTR 33.3%.	Explicit	IoT	Text (history + prompts)
RS-Stega: A Token-Level Quality-Control Framewo...	Generative language model with token-level quality controller	2025.0	Three benchmark datasets	Reports a 52.22% reduction in PPL and a 49.63% increase in BPW compared with baselines.	Implicit	Token context	Token probabilities and quality-control decisions